Task 5 – Redundancy & Data Migration

Objective

To understand Azure Storage redundancy options (LRS, ZRS, GRS, and RA-GRS), compare their differences and use cases, and select a suitable method for migrating a large on-premises dataset to Azure.

1. Azure Storage Redundancy

Azure Storage redundancy means storing multiple copies of data to protect it against hardware failures, datacenter failures, and regional disasters. It improves data durability and availability.

1.1 LRS (Locally Redundant Storage)

LRS stores three copies of data within a single physical location in one Azure region. It protects data against local hardware failures but does not protect against the failure of the entire datacenter or region.

Use Case: Development environments, testing, and non-critical data where cost is a priority.

1.2 ZRS (Zone-Redundant Storage)

ZRS replicates data across multiple availability zones within the same Azure region. It protects data against the failure of an individual availability zone while keeping data within the same region.

Use Case: Applications that require high availability and protection against zone failures.

1.3 GRS (Geo-Redundant Storage)

GRS stores three copies of data in the primary region and asynchronously replicates data to a secondary region. It protects data against regional disasters. The secondary data is not normally available for direct reading unless a failover occurs.

Use Case: Business-critical applications, backups, and disaster recovery.

1.4 RA-GRS (Read-Access Geo-Redundant Storage)

RA-GRS provides the same geo-replication as GRS but also allows read-only access to data in the secondary region. Applications can read replicated data through a secondary endpoint without waiting for failover, subject to replication delay and service availability.

Use Case: Applications and reporting systems that require read access to secondary-region data during a primary-region outage.

2. Comparison of Redundancy Options

Feature	LRS	ZRS	GRS	RA-GRS
Data replication	Single location	Multiple availability zones	Primary and secondary regions	Primary and secondary regions
Hardware failure protection	Yes	Yes	Yes	Yes
Zone failure protection	No	Yes	Depends on primary-region configuration	Depends on primary-region configuration
Regional disaster protection	No	No	Yes	Yes
Read access to secondary region	No	No	After failover	Yes, read-only
Relative cost	Low	Higher than LRS	Higher than LRS	Generally higher than GRS

3. Data Migration

Data migration is the process of transferring data from one storage system or location to another, such as from an on-premises datacenter to Azure Blob Storage.

Azure provides different tools for data migration depending on the dataset size, network speed, cost, and transfer requirements.

3.1 AzCopy

AzCopy is a command-line utility used to transfer files and data between local systems and Azure Storage over a network connection.

Advantages:

- Suitable for online data transfers.
- Supports large files and folders.

- Supports recursive copying and transfer job logs.
- Can resume interrupted transfers.
- Can be automated using scripts.

Suitable Data Size:

- Less than 100 GB: Suitable for quick online transfers.
- 100 GB to 1 TB: Suitable for regular data migration over a reliable network.

These ranges are practical guidelines, not fixed Azure limits. Actual transfer time depends on network speed, file sizes, and other conditions.

Use Case: AzCopy is suitable when a reliable internet connection is available and data needs to be transferred directly to Azure Storage.

Example Command:

```
azcopy copy "D:\OnPremData\*" "https://<storage-
account>.blob.core.windows.net/<container>" --recursive
```

The local path, storage account name, and container name must be replaced with actual values. Appropriate authentication and permissions are required.

3.2 Azure Import/Export Service

Azure Import/Export Service allows organizations to transfer large amounts of data using physical disk drives shipped to an Azure datacenter. It is useful when transferring data over the internet would take too long.

Advantages:

- Suitable for large datasets.
- Reduces dependence on internet bandwidth for the bulk transfer.
- Useful for organizations with slow or limited network connections.
- Requires compatible physical drives, proper preparation, and an import job.

Suitable Data Size:

- Several TBs: Worth considering when online transfer is slow or impractical.
- 10 TB or more: Can be a suitable option when network bandwidth is limited.
- Smaller datasets may also use physical transfer if network connectivity is poor or offline transfer is required.

These are approximate planning guidelines, not fixed Azure service thresholds. Physical disk capacity, supported workflows, shipping costs, and processing time should be checked before selecting this method.

Use Case: Import/Export is suitable when an organization needs to transfer terabytes of data and online transfer is impractical.

4. Comparison of AzCopy and Import/Export Service

Feature	AzCopy	Import/Export Service
Transfer method	Network/internet	Physical disk shipment
Suitable data size	GBs to multiple TBs	Usually large datasets of multiple TBs
Internet requirement	Reliable bandwidth is important	Not required for the bulk physical transfer
Hardware requirement	Computer or server	Compatible disk drives
Automation	Supports scripting	Requires physical preparation and shipment
Transfer time	Depends on network speed	Depends on copying, shipping, and processing
Main limitation	Network speed and transfer duration	Shipping time and operational complexity

5. Scenario-Based Decision

Scenario: A company wants to migrate 20 TB of data from its on-premises datacenter to Azure Blob Storage. The company has a slow internet connection, and transferring the complete dataset online would take several days.

Selected Method: Azure Import/Export Service

Reason:

1. The dataset is very large (20 TB).
2. The available internet bandwidth is limited.
3. Physical disk transfer can reduce dependence on network bandwidth.
4. The company can copy data to compatible drives and ship them to Azure for import.
5. The company must consider shipping time, drive compatibility, and service availability before proceeding.

If the company has a fast and reliable internet connection, AzCopy would be a suitable option because it supports direct online transfers and automation.

AzCopy vs Import/Export Service

Data Size	Suitable Method	Reason
Less than 100 GB	AzCopy	Easy and fast over a good internet connection.
100 GB – 1 TB	AzCopy	Suitable for regular online data transfers.
1 TB – 10 TB	AzCopy or Import/Export	Depends on internet speed and transfer time.
More than 10 TB	Import/Export or physical data transfer	Useful when network bandwidth is limited.

Result

Studied Azure Storage redundancy options, including LRS, ZRS, GRS, and RA-GRS. Compared their features, availability, disaster recovery capabilities, and use cases. Also compared AzCopy and Azure Import/Export Service and selected a suitable migration method for a large on-premises dataset.

Conclusion

Azure Storage redundancy protects data against different types of failures. LRS provides local protection, ZRS protects against zone failures, GRS provides regional disaster recovery, and RA-GRS additionally supports read access to secondary-region data.

AzCopy is suitable for online transfers over a reliable network, from smaller datasets to multiple terabytes. Azure Import/Export Service is useful for large datasets when internet bandwidth is limited. The appropriate solution depends on data size, network availability, transfer deadlines, recovery requirements, and cost.